ARXIV 2607.02668

# Improving LLMs via Validator-to-Generator Alignment

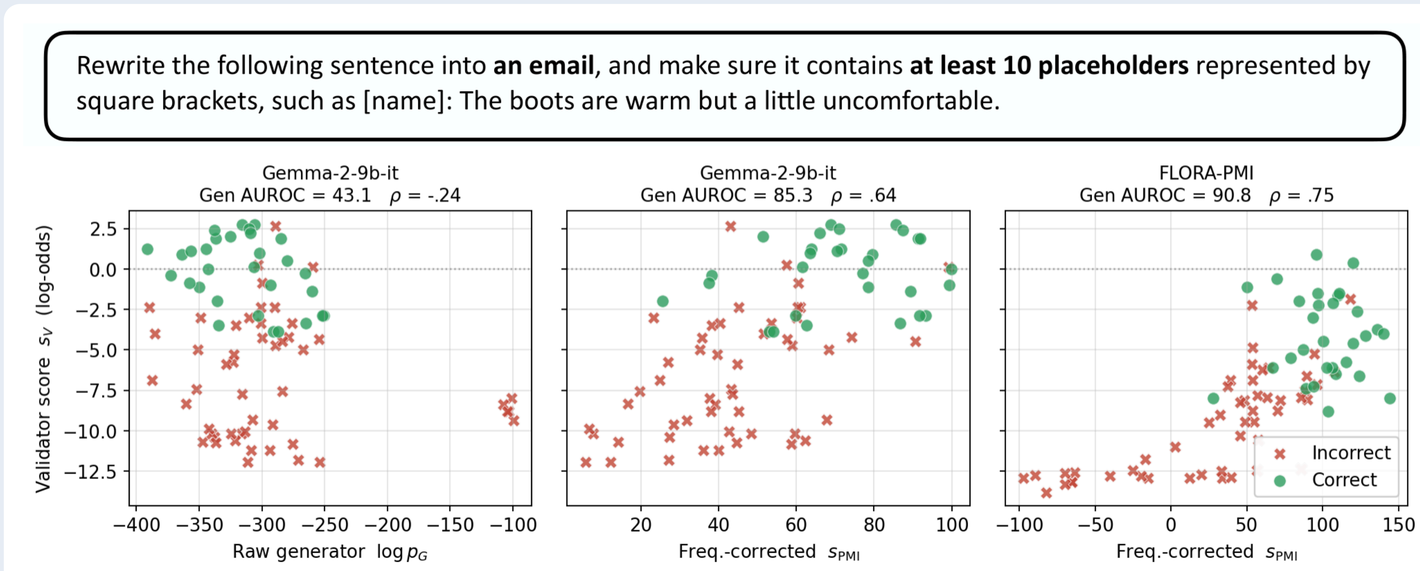
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**Paper**
arxiv.org/abs/2607.02668

**Code and data**
github.com/juand-r/flora

# Correct for how often an answer is said, and an LLM's generator lines up with its own validator.




We derive the correction from a model of a rational agent, then train on it. FLORA raises generator–validator correlation by up to 0.27 and generator AUROC by up to 7.3 points over the strongest alternative, without hurting validator AUROC.

One IFEval prompt, Gemma-2-9b-it. Each point is a candidate answer (green: correct, red: incorrect). The vertical axis is the model's own validator.

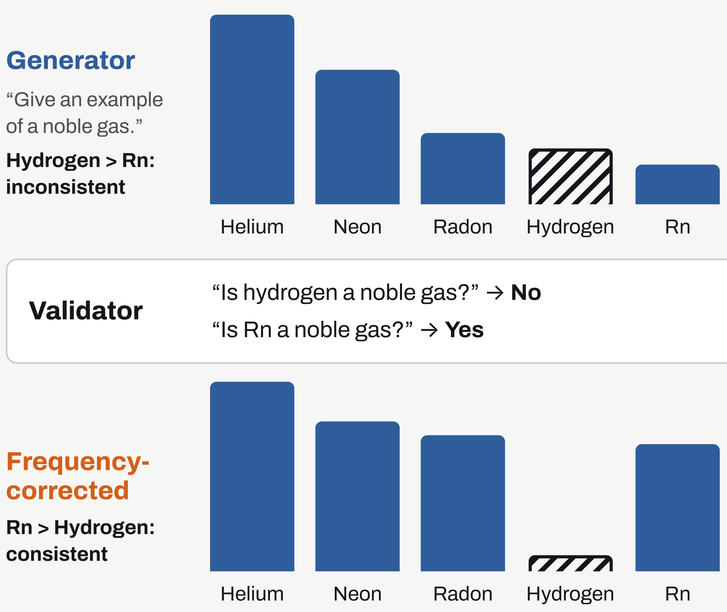
**Left, raw score.** No relation to the validator (ρ = −0.24).

**Middle, corrected, no training.** Most of the agreement appears (ρ = 0.64).

**Right, after FLORA.** Tighter still (ρ = 0.75).

## 1 Generators and validators disagree

Ask an LLM for an example of a noble gas and it rarely says "Rn". Ask it whether Rn is a noble gas and it says yes. The generator mixes up two things: whether an answer is correct, and how often people say it.




Illustration, after Figure 1 of the paper. Striped bar: the wrong answer.

## 2 What should agree with what

Model the LLM as an agent that is unsure which answers are correct. Then, for an answer $y$ to a prompt $x$, the validator's log-odds equal a frequency-corrected generator score:

$$\log \frac{p_V}{1-p_V} = \log p_G - \log p_{G'} + \log r$$

| validator log-odds | generator, asked for a correct answer | generator, asked for a wrong answer: how common $y$ is | assumed ≈ 0 |
|---|---|---|---|

So the raw generator score need not match the validator. The corrected score should.

## 3 FLORA: train for agreement

Frequency-corrected Learning of Ordered Rank Alignment turns this relation into a fine-tuning loss.

**Corrected generator score**
$s = \log p_G(y \mid x) - \log p_{G'}(y \mid x)$

**Validator log-odds**
from "Is $y$ correct?"

**Pairwise ranking loss**
Take two answers to the same prompt. The validator says which is better. Train the corrected generator score to rank them the same way.

FLORA-Neg gets $p_{G'}$ from a wrong-answer prompt. FLORA-PMI uses the unconditional $\log p(y)$ and needs no extra prompt. Likelihood terms on labeled examples keep both modes accurate.

### The correction matters during training

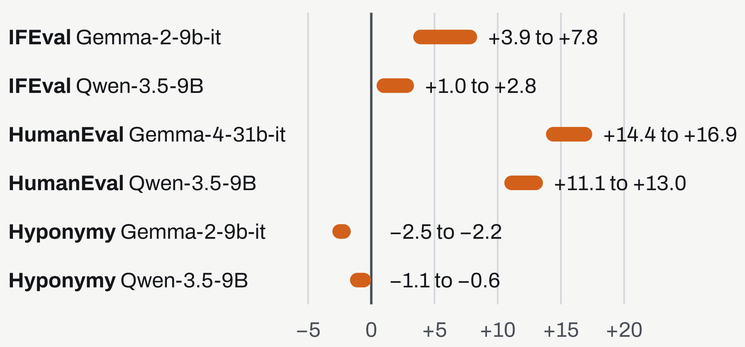
FLORA with and without frequency correction in its training loss. Both use the same correction at test time.

| Task | Gen. AUROC | ρ ×100 |
|---|---|---|
| IFEval Gemma-2-9b-it | 78.5 → **85.1** | 61.1 → **62.8** |
| HumanEval Gemma-4-31b-it | 87.9 → **94.2** | 60.2 → **76.3** |
| Hyponymy Gemma-2-9b-it | 90.9 → **92.5** | 73.1 → **73.8** |

Without → with the correction in training.

### The validator does not pay for it

Change in validator accuracy after FLORA training, in points. Each bar spans the two FLORA variants.

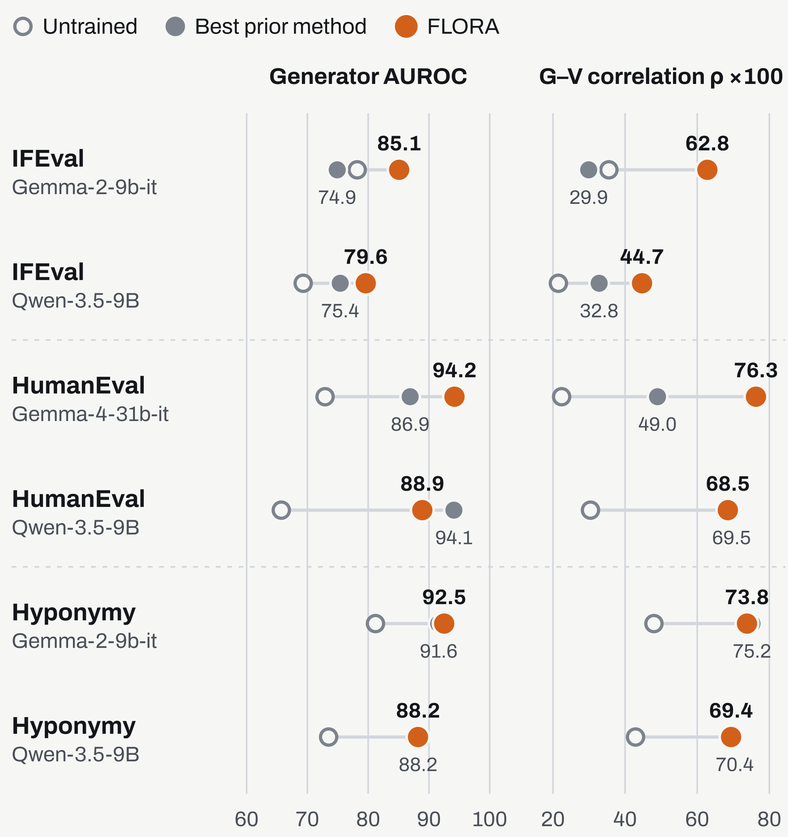



Validator accuracy rises on IFEval and HumanEval and dips by at most 2.5 points on Hyponymy. Validator AUROC never drops by more than 0.4 points.

## 4 Results

Three tasks: instruction following (IFEval), Python coding (HumanEval), category membership (Hyponymy). **Generator AUROC**: how well the corrected generator score separates correct from incorrect answers. **ρ**: Pearson correlation between generator score and validator log-odds. Both per prompt, then averaged.

### Better generators, closer agreement




FLORA is best or tied on generator AUROC in 5 of 6 settings, and best on ρ in 3 of 6 (within 1.4 points elsewhere). Its one clear loss is HumanEval with Qwen, where SFT reaches 94.1 AUROC to FLORA's 88.9.

Each dot is the best variant per setting and metric: FLORA-PMI or FLORA-Neg for FLORA; SFT, Consistency FT or RankAlign for prior methods. "Strongest alternative" above also includes the untrained model.

### Limits

The theory needs $\log r \approx 0$, which cannot be measured. Gains are in ranking fixed candidate sets; sampled and greedy outputs stay roughly unchanged. Correctness is treated as binary.